

QSVM Track — Week 5

Four-model baselines, kernel diagnostics, and the first significance tests

2 August 2026

Week 5 in one slide

Three results change conclusions this track had already drawn. They do not merely add to them.

Position entering the week	Now	Basis
Classical comparison is SVM only	Resolved	Four models, both tasks
CIC 15-class collapse unexplained	Attributed	Kernel concentration vs. an RBF control
No significance testing	Resolved	84 paired tests, all $p < 0.05$
Subsample fidelity assumed	Verified	KS rejections far below chance
SOREL blocked	Closed	Resume conditions recorded

Takeaway: the classical–quantum gap is real, larger than we had reported, and we now know *why* it appears on one dataset and not the other.

Data and protocol

CIC-MalMem-2022 — memory dumps, 55 forensic features

- binary: 58 596 rows, balanced
- 15-class: 29 298 malware rows, $1.7\times$ imbalanced

EMBER 2018 — static PE, 2381 features

- binary: 18 014 rows, 88.9% malware
- 15-class: 14 325 rows, 955 per family (balanced)

Metric: macro-F1, not accuracy. On EMBER binary at $n_c = 1$ the QSVM reaches 0.890 accuracy in 3 of 5 folds while benign-class F1 is exactly 0.000 — 88.9% of rows are malware and the model labels everything malware.

Fair-comparison protocol

- quantum kernel costs $O(n^2)$ circuit evaluations $\Rightarrow$ stratified 1000-row subsample
- that subsample and its 5 outer folds are *persisted and replayed* by the classical runner
- one shared pipeline: variance filter $\rightarrow$ correlation filter $\rightarrow$ standardise $\rightarrow$ PCA to n_c
- n_c fixes classical feature count *and* qubit budget at once
- fit on training-fold rows only

Result 1 — the classical baseline was too weak

Every EMBER margin on record compared the QSVM against *SVM alone*. Adding random forest, XGBoost and LightGBM:

	Random forest	XGBoost	LightGBM	SVM	QSVM angle	QSVM IQP
<i>Binary</i>						
$n_c = 1$	0.5582	0.5451	0.5093	0.5314	0.4532	0.4587
$n_c = 3$	0.6575	0.6465	0.6434	0.6306	0.4924	0.5084
$n_c = 6$	0.6835	0.6680	0.6593	0.6724	0.5138	0.5137
<i>15-class</i>						
$n_c = 1$	0.5562	0.5313	0.5410	0.5294	0.1688	0.1541
$n_c = 3$	0.7194	0.7089	0.7035	0.7389	0.3908	0.4797
$n_c = 6$	0.7903	0.7659	0.7709	0.7930	0.4232	0.5195

Takeaway: random forest beats SVM on every binary cell — so the margins we had published were measured against a sub-optimal baseline.

The margins we reported were understated

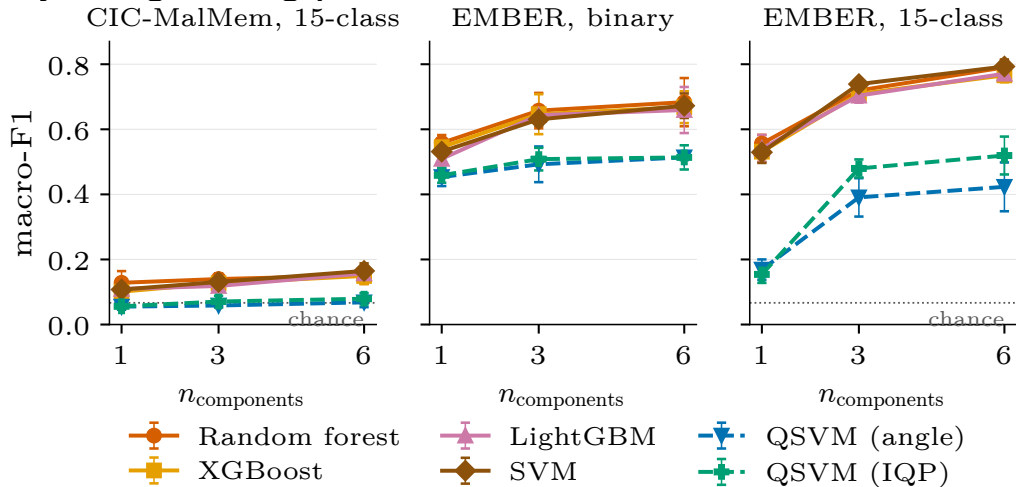
	$n_c = 1$	$n_c = 3$	$n_c = 6$
gap vs. SVM (as previously reported)	+0.0727	+0.1222	+0.1587
gap vs. <i>best</i> classical	+0.0995	+0.1491	+0.1697
understated by	0.0268	0.0269	0.0110

On the 15-class task SVM genuinely *is* the strongest baseline at $n_c \in \{3, 6\}$, so those two margins stand unchanged.

Four of six cells move. All four move the same way.

Takeaway: the correction goes *against* the quantum pipeline everywhere. Our earlier numbers flattered it.

Capacity scaling — the gap widens, it does not close



Takeaway: on EMBER binary the QSV gains 0.055 macro-F1 from 1 to 6 components; the random forest gains 0.125. More qubits are not buying the quantum model the headroom the classical model extracts from the same components.

The open question we inherited

On **CIC 15-class** the QSVM sits at the random baseline (0.056–0.079 vs. chance ≈ 0.067).

On **EMBER 15-class** the same method reaches 0.42–0.52.

Already ruled out: class imbalance, qubit count.

Why?

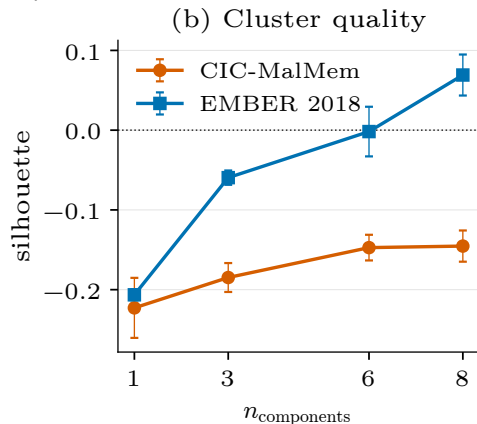
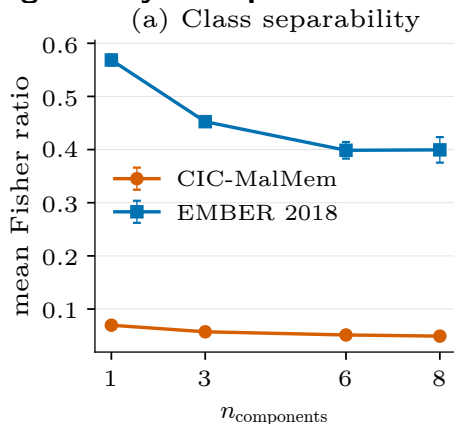
Takeaway: whatever CIC's features lack, the classical kernel finds something in them that the fidelity kernel does not. So (1) alone cannot be the answer.

Two candidate explanations, and they are testable separately:

1. the **features** are harder on CIC
2. the **kernel** fails on CIC

A classical SVM extracts 0.108–0.165 from *the same projected features* where the QSVM extracts 0.056–0.079.

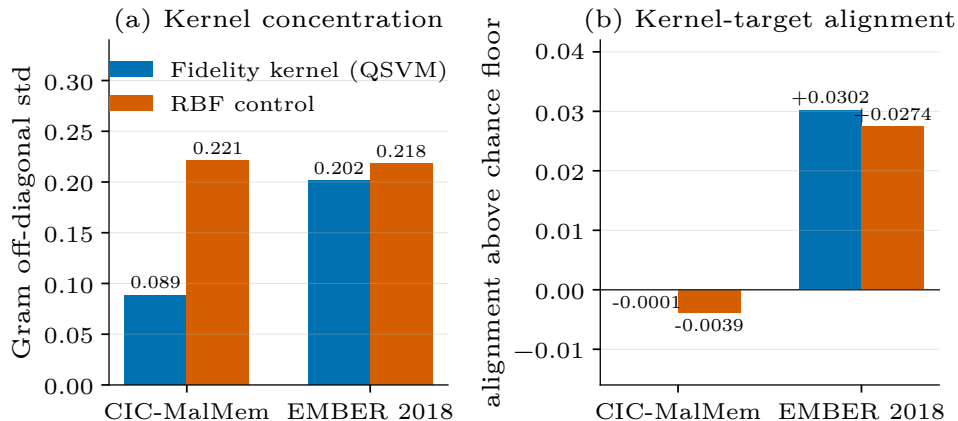
Feature geometry — explains the dataset, not the kernel



EMBER's families are $\approx 8\times$ more separable by Fisher ratio, up to $50\times$ on the worst-case family. CIC's silhouette is **negative at every** n_c : the average point lies closer to another family's centroid than its own. CIC's PCA nonetheless explains *more* variance (33–81% vs. 8–30%) — its variance sits in directions that do not discriminate family.

Takeaway: this explains why *every* method does badly on CIC 15-class. It cannot explain the QSVM-specific failure.

Kernel geometry — the answer



Both panels: first outer training fold, $n_c = 6$, IQP encoding, 800 rows. The RBF control is built on the **identical** feature matrix — that is what separates “these features are hard” from “this kernel is bad”.

Takeaway: RBF produces the same spread on both datasets (0.221 vs. 0.218). The fidelity kernel matches it on EMBER (0.92 $\times$) and **collapses to 0.40 $\times$ on CIC**.

What the kernel result does and does not support

Supported

The failure on CIC is a property of **the kernel**, not of the dataset. A classical kernel on identical features does not concentrate; the quantum one does. The Gram approaches a scaled identity — every pair looks equally similar — and an SVM given that has nothing to separate on. That is what a macro-F1 pinned to chance looks like from the inside.

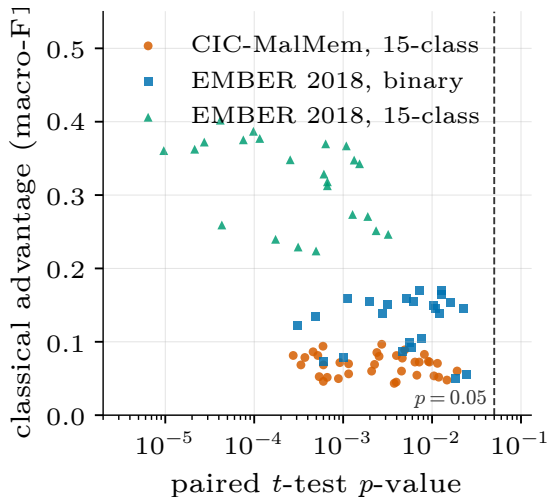
Not supported

That tuning `bandwidth` would repair it. Bandwidth is the identified mechanism and has never been tuned on CIC beyond its default — but no sweep was run. This is a hypothesis with a named lever, not a measured fix.

One tension we state rather than resolve

Alignment says neither kernel's Gram carries label information on CIC, yet a classical SVM reaches 0.16 there. Alignment is a global unweighted average over all pairs; an SVM needs no such thing. It ranks dataset difficulty well and predicts achievable SVM performance poorly.

Result 3 — the first significance tests



Specified in Week 1. Built, unit-tested, and **never run** until now.

Four weeks of “QSVM loses to classical” rested on differences in mean macro-F1, with no test that any gap exceeded fold-to-fold noise.

All 84 testable pairs favour the classical model, and all 84 reach $p < 0.05$ on the paired t -test.

Worst p -value across every pair: **0.0241**.

Takeaway: the headline claim now has evidence behind it, not an unexamined mean difference.

Two things we will not overclaim

Wilcoxon returned 0.0625 for all 84 pairs — that is not a null result

With 5 paired samples, 0.0625 is the *smallest attainable* two-sided p -value. The test is mathematically incapable of reaching 0.05 at this fold count, whatever the effect size. All 84 hitting the floor means every pair achieved the most extreme rank configuration available. *Fix*: 6 folds would put 0.03125 in reach — not a different test.

All CIC binary comparisons are untestable

Every CIC binary run — classical **and** quantum — predates parent/child run tracking and carries no recoverable sweep boundary. Raw records show the QSVM at 0.92–1.00, consistent with the earlier “ties on binary” reading, but they resolve into two distinct sweeps per n_c with no identifier separating them. *We did not reconstruct a grouping*. Doing so would invent a boundary the data does not record. The limitation is symmetric across frameworks.

Also closed this week

Subsample representativeness — specified in Week 2, never executed

Every quantum result uses a 1000-row subsample. Read against its null expectation — a faithful subsample still produces $\approx \alpha \cdot n_{\text{features}}$ rejections by chance — CIC rejects **nothing** of 55 features where ≈ 2.75 were expected; EMBER rejects **one** of 2381 where ≈ 119 were.

Class-proportion drift < 0.001 . *Four weeks of results now rest on a measured foundation.*

SOREL-20M — closed, not deferred

The labelling scheme was designed and validated over 19.7M rows and remains delivered work; the feature store was never fetched and **no sweep was ever run**. The two are independent — the labelling work is not partial credit for the sweeps. Neither disk (227 GiB free vs. 71.6 GiB) nor memory is the blocker; it is the fixed 71.6 GiB transfer, because the store offers no key-level remote access.

Summary

- Classical models beat the QSVM on **every** testable configuration — now with $p < 0.05$ on all 84, not on mean differences alone.
- The EMBER margins were **understated**, not overstated. Every correction this week went against the quantum pipeline.
- The one configuration where the quantum pipeline was ever competitive — CIC binary — cannot be significance-tested from what was logged, and we say so rather than dropping it quietly.

The transferable result

The fidelity kernel does not fail on CIC because CIC's features are hard — a classical RBF kernel on the identical features does not concentrate. It fails because **the fidelity kernel collapses to $0.40\times$ the classical kernel's spread** on that data. That is a statement about the method, not about the dataset.

Next: tune bandwidth on CIC — the named mechanism, still untested; move to 6 folds so a non-parametric test becomes usable.